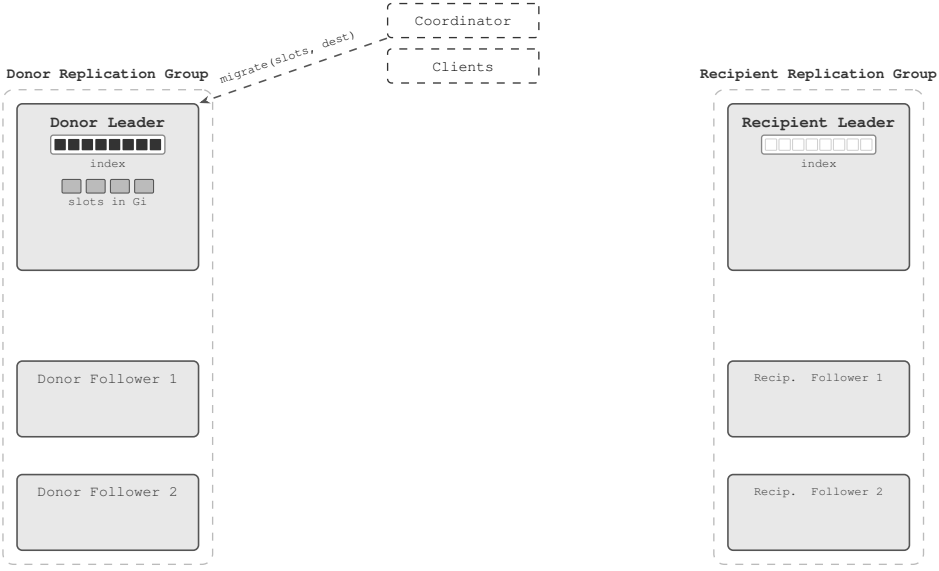
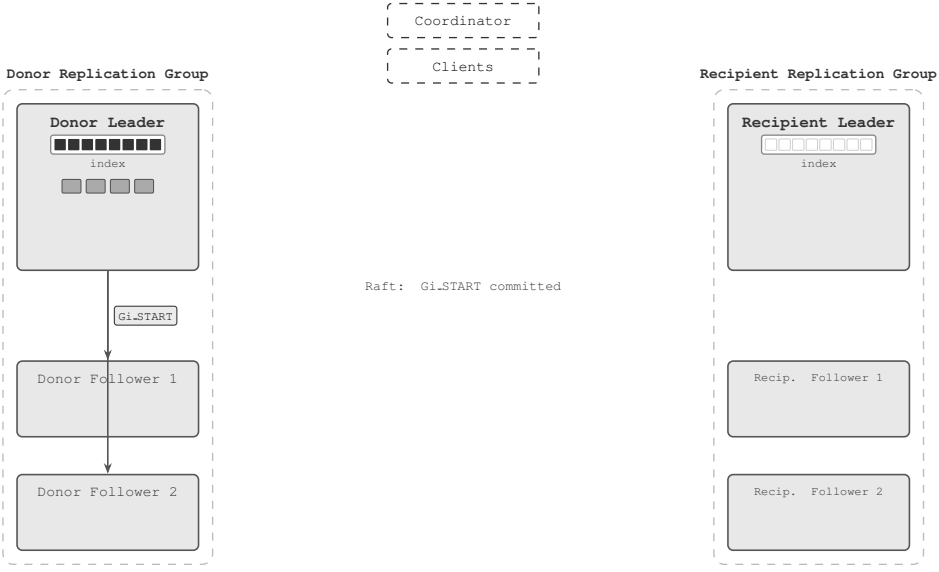
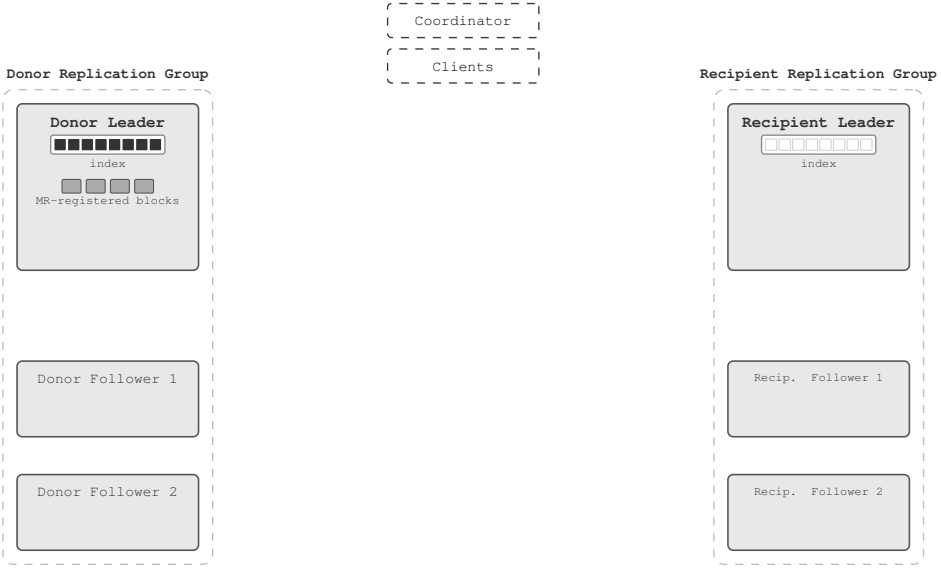
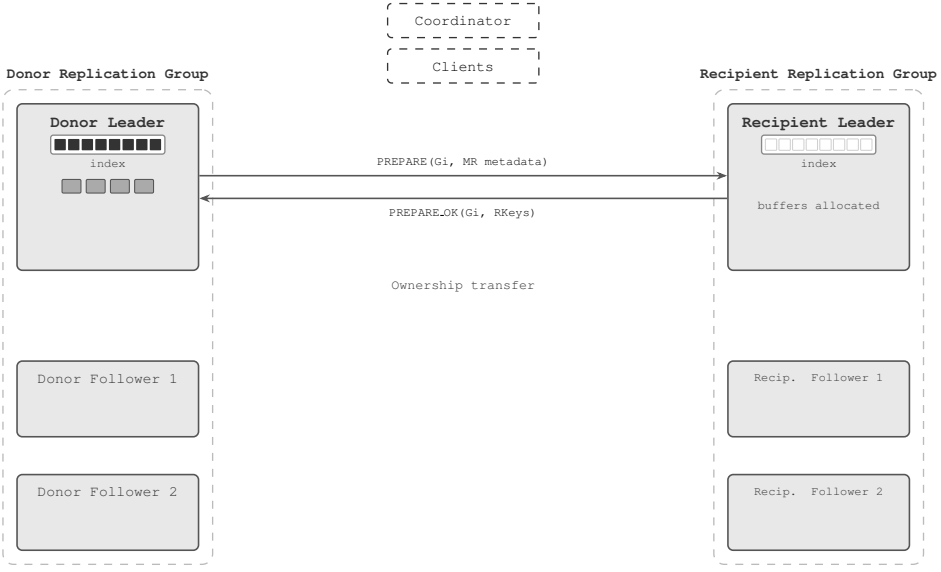
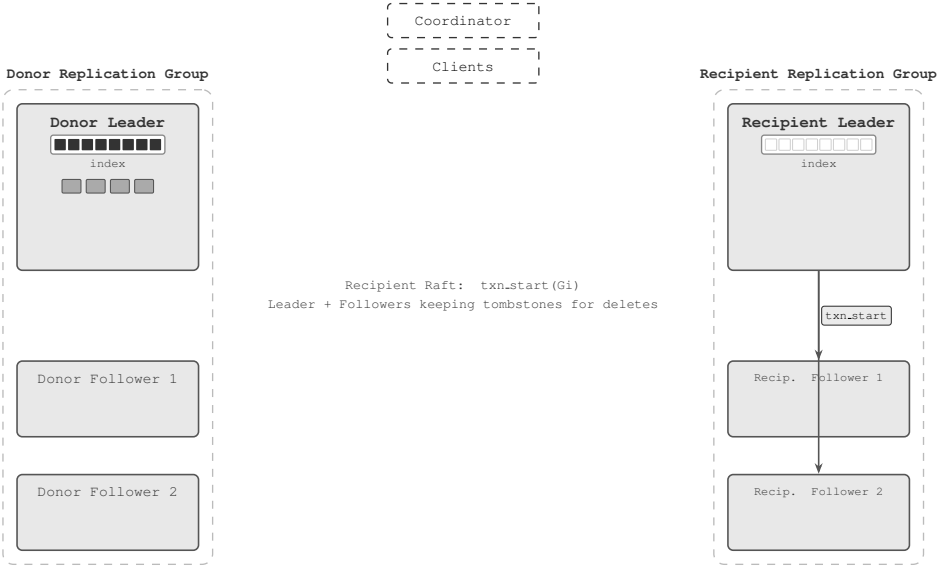
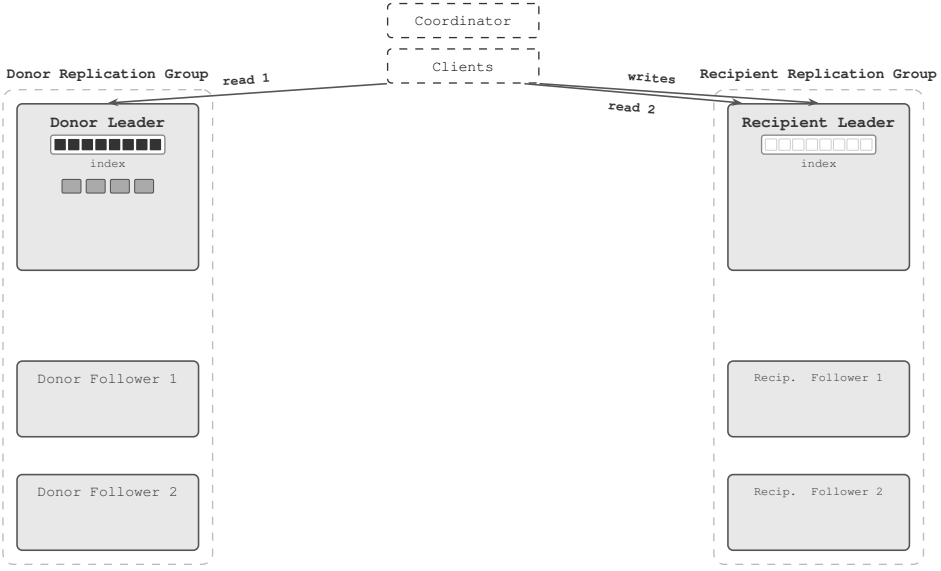

Coordinator
Clients
migrate(slots, dest)
Donor Replication Group
Donor Leader
index
slots in Gi
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2

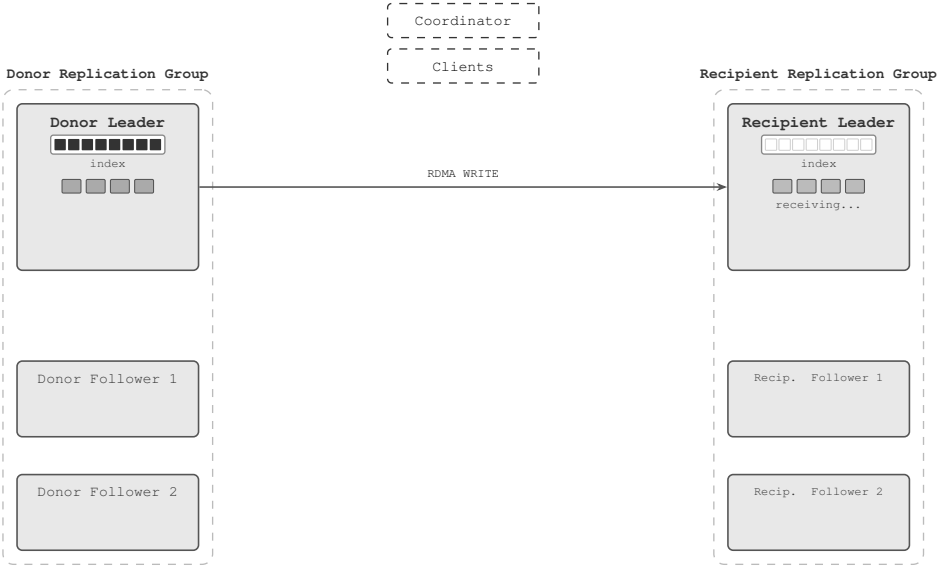

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Gi_START
Donor Follower 1
Donor Follower 2
Raft: Gi_START committed
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2

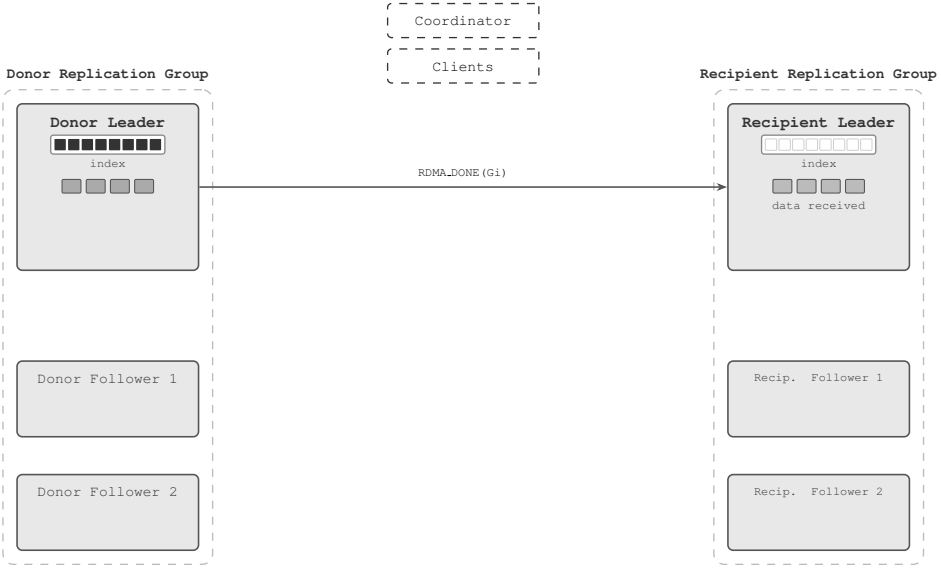

Coordinator
Clients
Donor Replication Group
Donor Leader
index
MR-registered blocks
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2

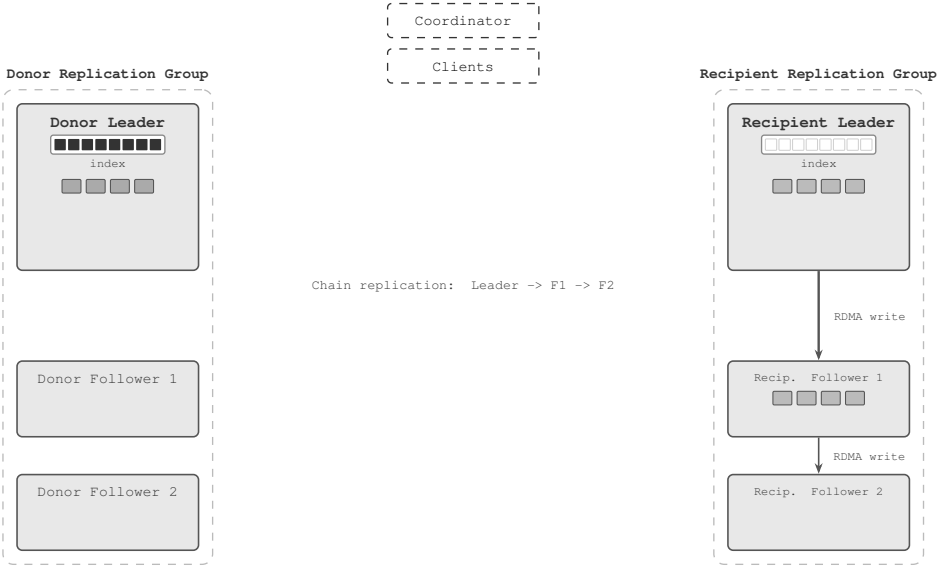

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
PREPARE(Gi, MR metadata)
PREPARE_OK(Gi, RKeys)
Ownership transfer
Recipient Replication Group
Recipient Leader
index
buffers allocated
Recip. Follower 1
Recip. Follower 2

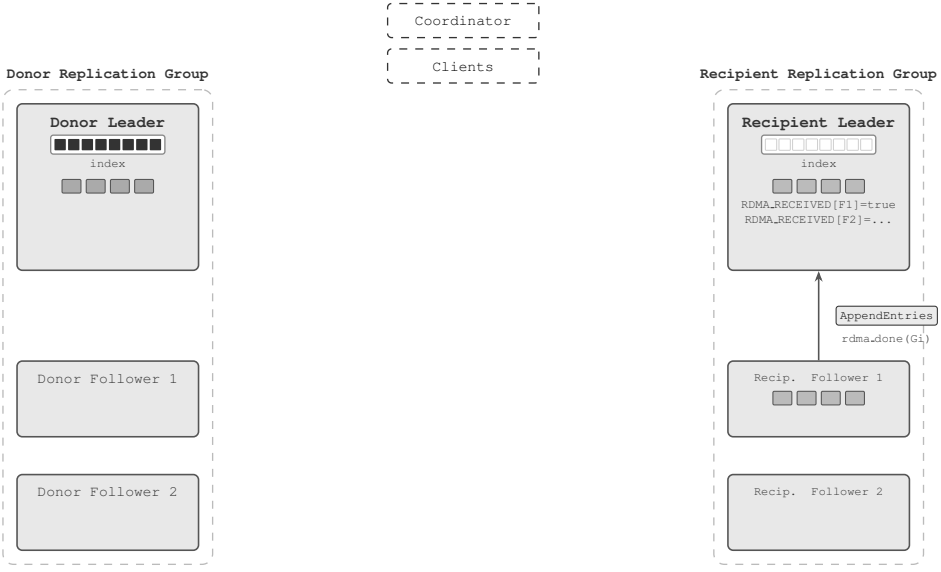

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Recipient Raft: txn_start(Gi)
Leader + Followers keeping tombstones for deletes
Recipient Replication Group
Recipient Leader
index
txn_start
Recip. Follower 1
Recip. Follower 2

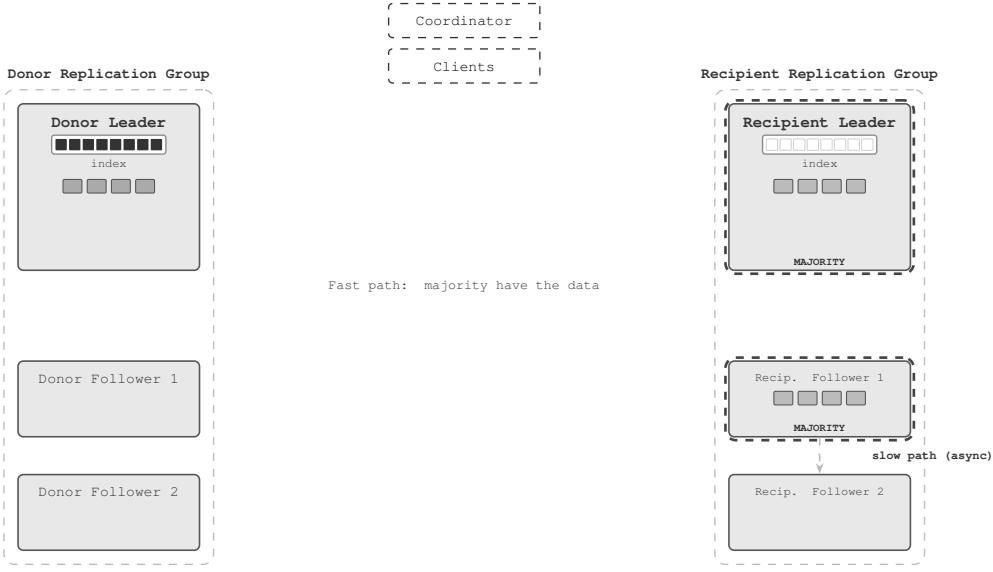

Coordinator
Clients
Donor Replication Group
read 1
writes
read 2
Recipient Replication Group
Donor Leader
index
Recipient Leader
index
Donor Follower 1
Donor Follower 2
Recip. Follower 1
Recip. Follower 2

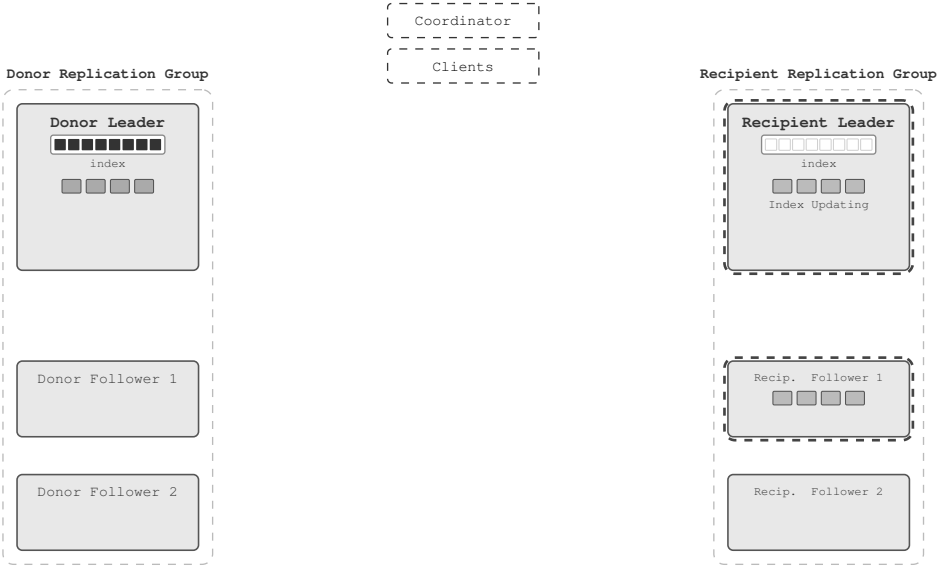

Coordinator
Clients
Donor Replication Group
Donor Leader
index
RDMA WRITE
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
receiving...
Recip. Follower 1
Recip. Follower 2

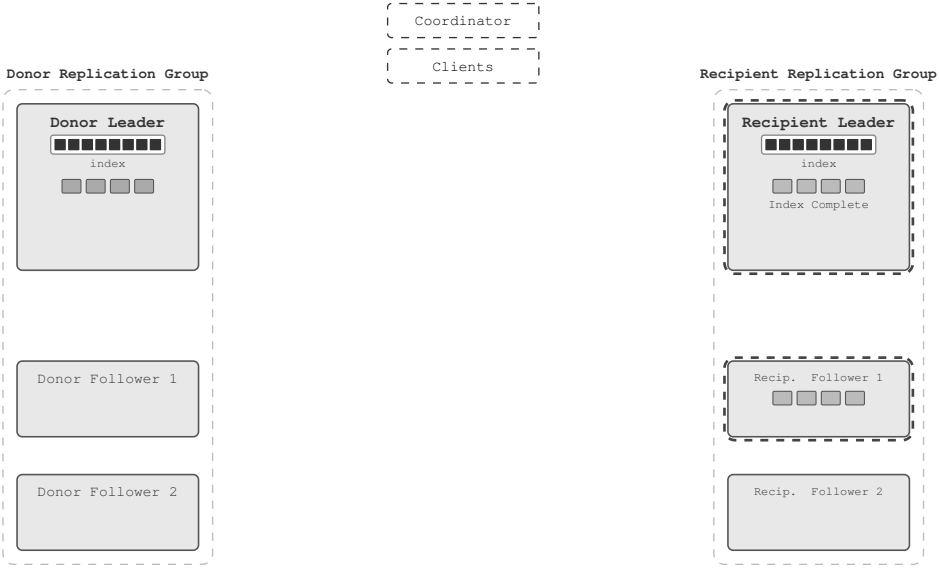

Coordinator
Clients
Donor Replication Group
Donor Leader
index
RDMA_DONE(Gi)
Recipient Replication Group
Recipient Leader
index
data received
Donor Follower 1
Donor Follower 2
Recip. Follower 1
Recip. Follower 2

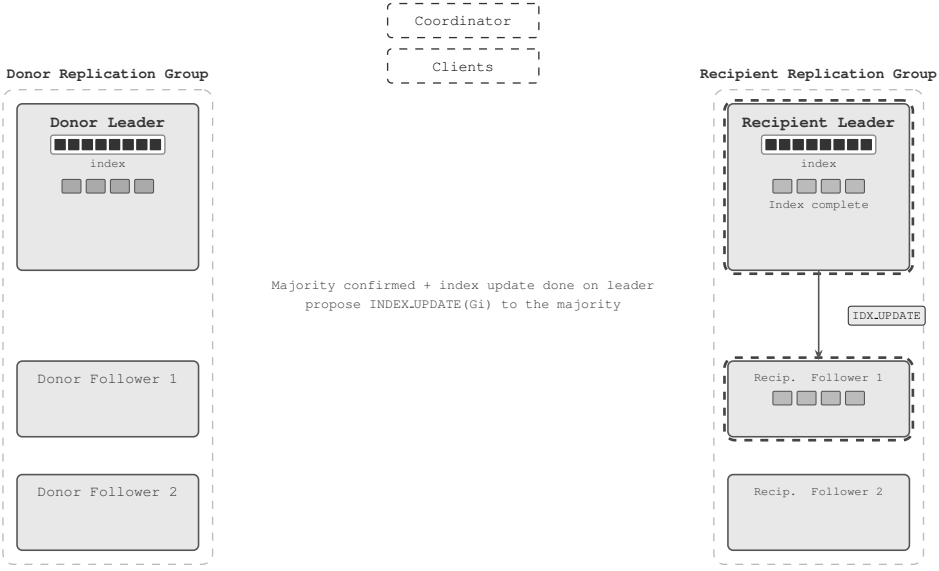

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Chain replication: Leader -> F1 -> F2
Recipient Replication Group
Recipient Leader
index
RDMA write
Recip. Follower 1
RDMA write
Recip. Follower 2

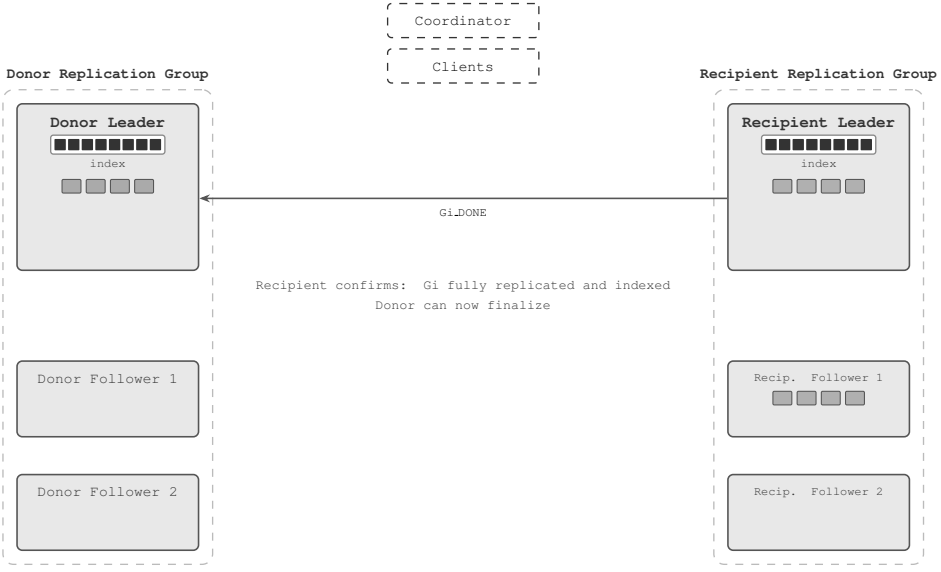

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
RDMA_RECEIVED[F1]=true
RDMA_RECEIVED[F2]=...
AppendEntries
rdma_done(Gi)
Recip. Follower 1
Recip. Follower 2

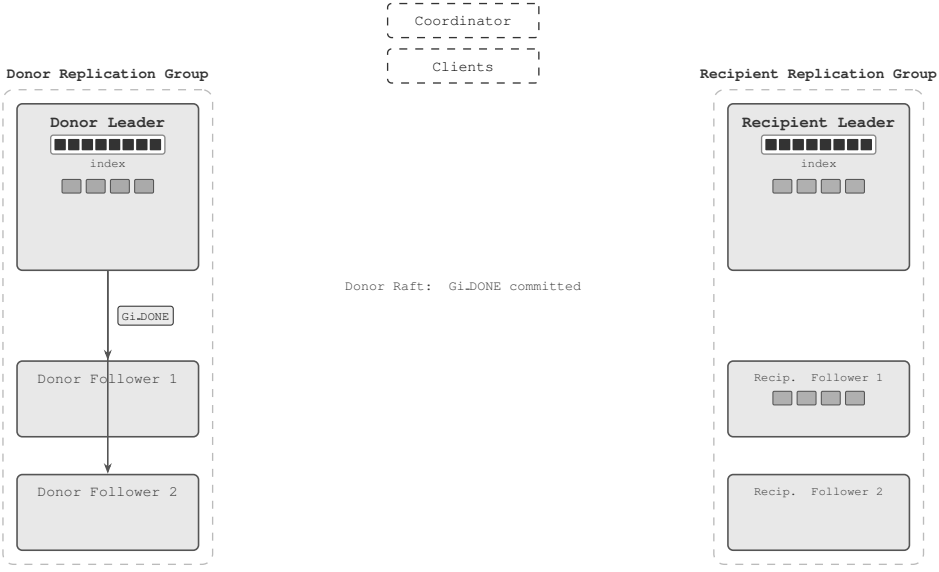

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Fast path: majority have the data
Recipient Replication Group
Recipient Leader
index
MAJORITY
Recip. Follower 1
MAJORITY
slow path (async)
Recip. Follower 2

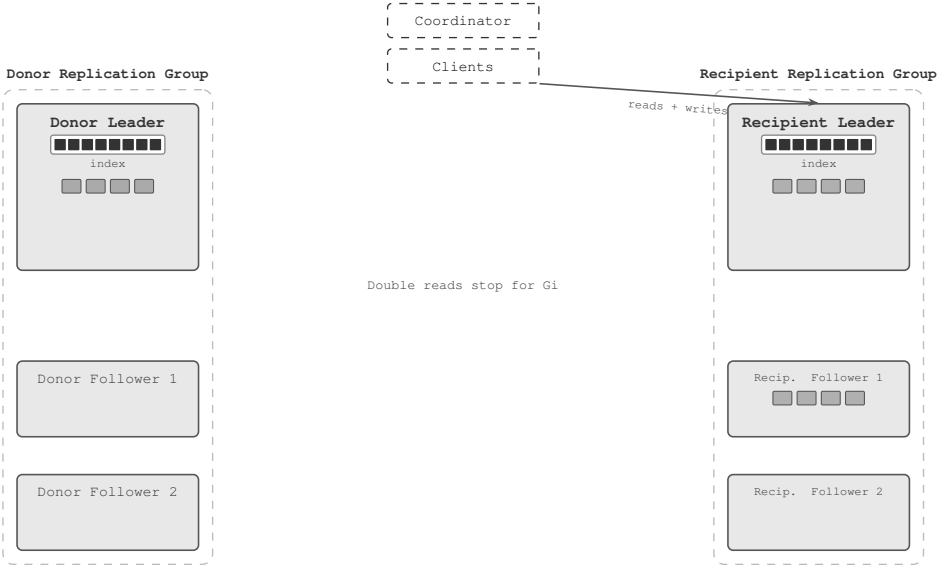

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Index Updating
Recip. Follower 1
Recip. Follower 2

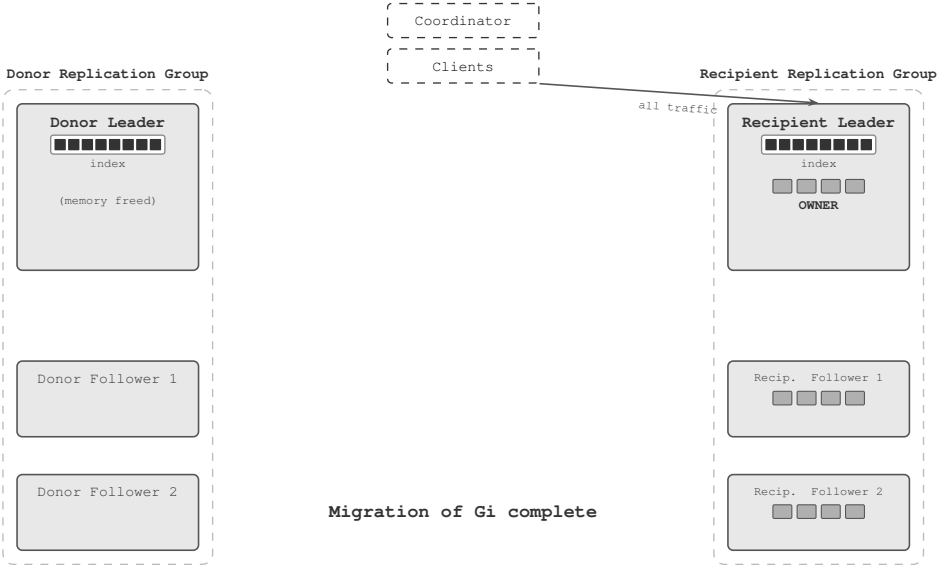

Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Index Complete
Recip. Follower 1
Recip. Follower 2


Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Majority confirmed + index update done on leader
propose INDEX_UPDATE(Gi) to the majority
Recipient Replication Group
Recipient Leader
index
Index complete
IDX_UPDATE
Recip. Follower 1
Recip. Follower 2


Coordinator
Clients
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2
Gi_DONE
Recipient confirms: Gi fully replicated and indexed
Donor can now finalize


Coordinator
Clients
Donor Replication Group
Donor Leader
index
Gi_DONE
Donor Follower 1
Donor Follower 2
Donor Raft: Gi_DONE committed
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2


Coordinator
Clients
reads + writes
Donor Replication Group
Donor Leader
index
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2
Double reads stop for Gi


Coordinator
Clients
all traffic
Donor Replication Group
Donor Leader
index
(memory freed)
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
OWNER
Recip. Follower 1
Recip. Follower 2
Migration of Gi complete